Router and Switch CLI Test Report

Unity TSS2 training simulator | Play Mode validation | 16 October 2026

Executive Summary

The router and switch were installed in the Balanced_12_PIECE_LAB scenario and configured through the in-simulator CLI. The router configuration path reached Privileged EXEC with the expected hostname, route state, and saved startup configuration. After a clean Play Mode restart, the switch accepted both trunk commands; Fa0/1 became an 802.1Q trunk carrying the expected VLANs.

Test Environment

Item	Value
Application	Unity TSS2 training simulator
Scene	Balanced_12_PIECE_LAB
Execution	Unity Play Mode
Project changes	Added trunk-state regression assertions; no scene or prefab changes
Console result	0 errors and 0 warnings after testing

Router Test

Installed device: router at Rack01 U3. Runtime session key: Rack:Rack01:U03. Final hostname: CN-1.

Router Commands and Effects

Command	Runtime state effect	Result
enable	User EXEC to Privileged EXEC	Accepted
configure terminal	Privileged EXEC to Global Configuration	Accepted
hostname CN-1	Changed device hostname to CN-1	Accepted
no ip domain-lookup	Disabled DNS lookup	Accepted
service password-encryption	Enabled password encryption setting	Accepted
enable secret training	Configured the enable secret	Accepted
interface Fa0/0	Selected the physical router interface	Accepted
no shutdown	Enabled the selected interface	Accepted
interface Fa0/0.<VLAN>	Selected router-on-a-stick subinterfaces	Accepted during test

Command	Runtime state effect	Result
encapsulation dot1q <VLAN>	Applied 802.1Q VLAN tagging to subinterfaces	Supported path exercised
ip address <address> <mask>	Applied interface addressing	Supported path exercised
ip route ...	Configured a route; one route was present in final state	Accepted
show running-config / show ip route	Read-only inspection and evidence capture	Executed
copy running-config startup-config	Saved running configuration to startup state	Saved

Router Final State

- Hostname: CN-1
- Mode: Privileged EXEC
- DNS lookup: disabled
- Routes: one route present
- Startup configuration: saved
- Qualification: the router command path was exercised, but the session required explicit interface exits for reliable movement between interface contexts; subinterface retention was not treated as a clean pass in that run.

Switch Test

Installed device: Cisco 2960 PoE Switch at Rack02 U3. Runtime session key: Rack:Rack02:U03. Final hostname: AN-1.

Switch Commands and Effects

Command	Runtime state effect	Result
enable	User EXEC to Privileged EXEC	Accepted
configure terminal	Privileged EXEC to Global Configuration	Accepted
hostname AN-1	Changed device hostname to AN-1	Accepted
no ip domain-lookup	Disabled DNS lookup	Accepted
service password-encryption	Enabled password encryption setting	Accepted
enable secret training	Configured the enable secret	Accepted
vlan 80 / name MANAGEMENT	Created and named VLAN 80	Accepted
vlan 310 / name RECEPTION	Created and named VLAN 310	Accepted
vlan 20 / name DATA	Created and named VLAN 20	Accepted
vlan 190 / name NATIVE	Created and named VLAN 190	Accepted
interface Fa0/1	Selected the uplink interface	Accepted
switchport trunk encapsulation dot1q	Enabled the modeled 802.1Q encapsulation flag	Accepted; dot1q=True

Command	Runtime state effect	Result
switchport mode trunk	Changed Fa0/1 to trunk mode	Accepted; trunk=True
switchport trunk native vlan 190	Set native VLAN to 190	Accepted
switchport trunk allowed vlan 80,310,20,190	Set four allowed VLANs	Accepted
no shutdown	Enabled Fa0/1	Accepted
exit	Returned from interface configuration to Global Configuration	Required between interface selections
interface Fa0/2 + access VLAN 80	Enabled Fa0/2 and assigned VLAN 80	Accepted
interface Fa0/3 + access VLAN 310	Enabled Fa0/3 and assigned VLAN 310	Accepted
interface Fa0/4 + access VLAN 20	Enabled Fa0/4 and assigned VLAN 20	Accepted
end	Returned to Privileged EXEC	Accepted
show vlan brief	Read-only VLAN inspection and evidence capture	Executed
show interfaces trunk	Read-only trunk inspection and evidence capture	Executed
copy running-config startup-config	Saved running configuration to startup state	Saved

Switch Final State

Element	Final state
Hostname and mode	AN-1; Privileged EXEC
VLAN database	80 MANAGEMENT; 310 RECEPTION; 20 DATA; 190 NATIVE
Fa0/1	Administratively up; 802.1Q trunk; native VLAN 190; allowed VLANs 20, 80, 190, 310
Fa0/2	Administratively up; access VLAN 80
Fa0/3	Administratively up; access VLAN 310
Fa0/4	Administratively up; access VLAN 20
Startup configuration	Saved

Conclusion

The access-switch configuration, VLAN database, interface VLAN assignments, inspection commands, and startup save behavior are working in Play Mode. The rerun also confirmed that switchport trunk encapsulation dot1q and switchport mode trunk are accepted after a clean Play Mode restart. Fa0/1 reported 802.1Q trunking with native VLAN 190 and all four expected VLANs associated.